- Coordinates
- Coordinate mapping
- isomorphism

§4.4 Coordinate Systems

w/ bases we can identify certain v.s. w/ $\mathbb{R}^n$

Thm 8 (unique representation thm)

$\mathcal{B} = \{\vec{b}_1, \cdots, \vec{b}_n\}$ basis of v.s. V.

Then $\forall \vec{x} \in V$, $\exists!$ (there exist a unique) set of scalars $c_1, \cdots, c_n$ s.t.

$$\vec{x} = c_1\vec{b}_1 + \cdots + c_n\vec{b}_n$$

Def Coordinates of $\vec{x}$ relative to $\mathcal{B}$ are $c_1, \cdots, c_n$.

Coordinate vector $[\vec{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$

Exmp $\vec{x} = \begin{bmatrix} 1 \\ 6 \end{bmatrix} \in \mathbb{R}^2$ $\vec{e}_1, \vec{e}_2$ standard basis of $\mathbb{R}^2$

coordinates of $\vec{x}$ relative to $\{\vec{e}_1, \vec{e}_2\}$ are 1 & 6

Coordinate vector: $\begin{bmatrix} 1 \\ 6 \end{bmatrix}$

Exmp $\mathcal{B} = \left\{\vec{b}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \vec{b}_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}\right\}$ form a basis of $\mathbb{R}^2$ (why?)

$\vec{x} \in \mathbb{R}^2$ has coordinate vector $[\vec{x}]_{\mathcal{B}} = \begin{bmatrix} -2 \\ 3 \end{bmatrix}$, $\vec{x} = ?$ relative to $\mathcal{B}$

Sol $\vec{x} = -2\vec{b}_1 + 3\vec{b}_2 = -2 \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \end{bmatrix}$ ☆

Coordinate mapping V v.s. w/ basis $\mathcal{B} = \{\vec{b}_1, \cdots, \vec{b}_n\}$

The process of obtaining coordinates provides a mapping

$$[\ ]_{\mathcal{B}} : V \longrightarrow \mathbb{R}^n$$
$$\vec{x} \longmapsto [\vec{x}]_{\mathcal{B}}$$

Thm The mapping $[\ ]_{\mathcal{B}}$ is a linear transformation which is one-to-one & onto.

(It identifies the linear structures on both sides)

⇑ linearly ind. ⇑ span.

Def A lin. trans which is 1-1 & onto is an isomorphism ☆